Height Quotients of the Riemann Zero Divisor

Central-circle limits and invisible sparse perturbations

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Abstract

What survives when the heights of zeta zeros are folded around a fixed circle? We formulate that question for the Riemann zero divisor on a cylinder and on a half-twisted band. Established all-zero mod-one equidistribution, discrepancy, and zero-density estimates give a normalized central-circle limit and elementary support consequences. A known positive proportion of simple critical-line zeros gives packing and closure corollaries, without asserting equidistribution of that selected subsequence. We then give a concrete even entire product with dyadically sparse off-critical roots. Direct estimates transfer the stated quotient limits, known smooth prime-resonant secondary limits, and selected counting and Landau error classes. The normalized multiplier preserves the functional equation and pole residue but fails an ordinary Dirichlet-series condition. Prescribed zero insertion and functional-equation nonuniqueness have substantial prior work; this is a worked synthesis and an explicit example with quantitative controls, without a claim of priority. Three formula-based figures orient the construction.

1 Question, contribution, and arithmetic inputs

The guiding question is this: after a finite-height sample of zeros is folded by height modulo L, which features reflect the original divisor, and which features can a very sparse added divisor leave unchanged? The answer separates an exact obstruction from an averaged limit. Exact vertical repetition of the relevant support is impossible; nevertheless, after normalization, the folded samples approach the central circle, and a dyadically sparse off-critical addition can share the observations considered here.

The intended reader knows standard complex analysis and basic real analysis; we explain the measure vocabulary locally and state the zeta-zero inputs. A useful first route is Sections 3 and 6, for the folding, limit, and explicit example. Section 5 explains the finer smooth tests; Sections 2 and 4 give the exact-period and critical-line consequences. Section 7 identifies the arithmetic condition that the constructed function fails, and Section 8 suggests further questions. Proof details can then be read in order.

Scope. The paper concerns exact fixed-height periods of the specified support or divisor and the averaged tests below; it makes no assertion that individual gaps or approximate patterns never recur, and it neither proves nor refutes the Riemann hypothesis.

We use Riemann's entire function, with removable values understood,

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \xi(1-s) = \xi(s).$$

This is the normalization of [4, 25.4.3–25.4.4]; it also satisfies $\xi(\bar{s}) = \overline{\xi(s)}$. Center the critical strip by writing

$$\rho = \tfrac{1}{2} + x_\rho + i\gamma, \qquad -\tfrac{1}{2} < x_\rho < \tfrac{1}{2}.$$

One may picture the operation as folding the height axis around a circle of circumference L. This is a geometric folding of the ambient strip, not a claim that the marked zero divisor itself is periodic or descends to the quotient. A second distinction will organize the argument: pushing every positive-height zero forward gives an infinite raw mass, whereas restricting to $0 < \gamma \le H$ and dividing by its total multiplicity gives a probability measure. We make both constructions precise for a cylinder and a Möbius band.

The mod-one statistics used here are prior results of Hlawka and Fujii, recorded and developed by Ford–Zaharescu and Ford–Soundararajan–Zaharescu [6, 7]. The quotient and retraction arguments are convenient geometric formulations of their consequences, rather than a new zero-distribution theorem. Bui–Conrey–Young [5] supply the simple-critical-line proportion; our residue and closure bounds are packing and closed-set corollaries.

Earlier close perturbations of zeta were studied by Gauthier–Zeron [2].[1] More precisely, Gauthier–Xarles [1, Proposition 1 and Theorem 2] construct symmetric prescribed-zero perturbations retaining a Riemann-type functional equation. That work retains the old divisor away from the prescribed set and poles/principal parts, with strong approximation properties. Thus symmetric zero insertion itself is not a new idea. Our product is a specific dyadic order-zero example whose count, compact-tail and stated statistical transfers can be checked directly; we make no claim that its individual estimates are new analytic results. We do not obtain the strong function-value approximation of [1], nor infer our precise growth controls from its theorem.

A claim-targeted literature audit located no identical printed package of this product and all the listed transfer formulas. That bounded search result does not certify priority. The paper is presented as a worked synthesis and an explicit diagnostic obstruction, with the comparison in Table 1. Functions outside the ordinary Dirichlet-series class and converse-theorem boundaries have extensive prior literature [8, 3, 9].

Let m_ρ be the order of a nontrivial zero and put

$$N(H) = \sum_{0<\operatorname{Im}\rho\le H} m_\rho.$$

The *divisor* remembers each zero and its order m_ρ; the *support* remembers only locations. Every sum uses the divisor unless distinct support or simple zeros are explicitly specified. We use the following unconditional inputs, with their exact scopes recorded here.

1. The Riemann–von Mangoldt asymptotic is

$$N(H) = \frac{H}{2\pi}\log\frac{H}{2\pi} - \frac{H}{2\pi} + O(\log H). \tag{1}$$

[1] The publisher abstract and metadata were inspected for this historical comparison; the precise theorem comparison is to [1].

Table 1: Prior results and the limited role of the present formulation.

Prior work	Already established	Present role
Hlawka; Fujii; FZ/FSZ [6, 7]	All-zero circle statistics and smooth secondary limits	Quotient/retraction corollaries; no new distribution theorem
Bui–Conrey–Young [5]	A positive proportion of simple critical-line zeros	Elementary residue-packing and closure bounds
Gauthier–Xarles [1]	Symmetric zero insertion, functional equation, retained poles and approximation	Explicit dyadic product and simultaneous transfer calculations
Converse literature [8, 3, 9]	Dirichlet-series rigidity and constructions outside that setting	Direct real-axis obstruction for this multiplier

The count is finite in every bounded-height slab, since Riemann's entire function ξ is nonzero and its zeros are precisely the nontrivial zeta zeros, with orders.

2. For every fixed $\alpha \neq 0$, the fractional parts $\{\alpha\gamma\}$ of all nontrivial zero ordinates are uniformly distributed modulo one. Ford–Zaharescu [6, Introduction] state Hlawka's unconditional theorem and record Fujii's unconditional star-discrepancy bound

$$D_\alpha^*(H) = O_\alpha\left(\frac{\log\log H}{\log H}\right). \tag{2}$$

The original Hlawka and Fujii articles are identified in [6]; our direct source for these precisely stated consequences is that research paper.

3. If $N(\sigma, H)$ counts zeros with multiplicity, with $|\gamma| \leq H$ and $\operatorname{Re}\rho \geq \sigma$, Selberg's bound as stated in [6, Lemma 2] is, uniformly for $1/2 \leq \sigma \leq 1$,

$$N(\sigma, H) \ll H^{1-(\sigma-1/2)/4}\log H. \tag{3}$$

4. Bui–Conrey–Young [5, Theorem 1.1] prove

$$\liminf_{H\to\infty} \frac{N_0^*(H)}{N(H)} \geq 0.4058, \tag{4}$$

where $N_0^*(H)$ counts *simple critical-line* zeros of positive height. We use the conservative constant $c = 0.4$.

The secondary smooth-test theorem of [6] appears in Section 5; it is a theorem for fixed smooth tests, not an interval-indicator theorem. All inputs in this list concern actual zeta zeros. In particular, the all-zero equidistribution statement does not pass automatically to the critical-line-only subsequence.

We consistently use $0 < \gamma \leq H$. When a cited count uses a strict or averaged endpoint, squeeze it between nonordinate heights within distance one. The main term in (1) changes by $O(\log H)$ on such an interval; the same asymptotic class follows. For bounded smooth tests, any endpoint contribution to a $1/H$-scaled statistic is $O(\log H/H)$. In particular, changing the BCY endpoint count costs $O(\log H) = o(N(H))$, so its limiting proportion is unaffected.

2 Dense winding and support-period obstructions

We first ask what the folding cannot mean. Let S be the unweighted zero support in centered coordinates. For $L > 0$, write $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$ for the circle obtained by folding heights modulo L. Its normalized Haar probability m_L is simply uniform arc length: an arc I receives mass $|I|/L$. An eventual support period L means that, above some height H_0, membership is equivalent under $(x,t) \mapsto (x, t+L)$ whenever both heights are in that tail. Mere forward inclusion is a different hypothesis.

Theorem 2.1. *For every $L > 0$, every nonempty open arc of $\mathbb{T}_L$ contains infinitely many distinct residues of positive zero ordinates. The complete unweighted support has no full or eventual positive vertical period. It also has no full or eventual glide period $(x,t) \mapsto (-x, t+L)$.*

Proof. Uniform distribution gives, for every arc I,

$$\sum_{\substack{0<\gamma\leq H\\ [\gamma]_L\in I}} m_\rho = \left(\frac{|I|}{L} + o(1)\right) N(H). \tag{5}$$

In particular every open arc is met. If an open arc contained only finitely many distinct projected ordinates, choose a smaller nonempty open subarc avoiding them and apply (5) again. This proves the asserted infinitude, without discarding multiplicities from the equidistribution input.

Under eventual period equivalence, every sufficiently high zero can be translated downward into a slab of height L above H_0. There are only finitely many support points in that slab: embed it in a compact rectangle of the closed strip and use local finiteness of the entire ξ divisor. The finite positive-height prefix also has only finitely many points. All residues would thus belong to a finite set, contradicting the first assertion. Squaring an eventual glide gives an eventual vertical period $2L$, which is already excluded. □

For comparison, the simpler divisor argument only uses (1): periodicity of the *multiplicity divisor* gives the same finite mass in each half-open height cell, hence $N(H) = O(H)$, a contradiction. That argument does not by itself rule out periodic support with varying multiplicities. Theorem 2.1 adds the equidistribution input needed to address locations rather than weights.

Remark 2.2. Thus the conclusion is about exact fixed-height repetition of this support or divisor. It leaves open recurrence of individual gaps, approximate patterns, and periodic orbits in unrelated dynamical models.

3 Cylinder and Möbius probability limits

The preceding theorem concerns an exact divisor. We now ask what a large finite sample looks like after folding. The two quotients differ only in whether one trip around the height circle preserves or reverses the transverse coordinate.

Fix a closed width $A \geq 1/2$, a height length $L > 0$, and $e \in \{1, -1\}$. On $B_A = [-A, A] \times \mathbb{R}$ let $d_e(x, t) = (ex, t + L)$, and set $Q_e = B_A/\langle d_e \rangle$. The case $e = 1$ is a closed cylinder; $e = -1$ is a closed Möbius band. Denote the quotient map by q_e. The equivalence relation means the entire integer deck orbit, not just one identification.

Lemma 3.1. *The quotient topology is given by the metric*

$$d_{Q_e}([x, t], [y, u]) = \inf_{n \in \mathbb{Z}} \sqrt{(x - e^n y)^2 + (t - u - nL)^2}. \tag{6}$$

The space is compact. Its central embedding $j_e : \mathbb{T}_L \to Q_e$, $j_e([t]) = [0, t]$, is isometric. The retraction $r_e([x, t]) = [0, t]$ is well defined and satisfies

$$d_{Q_e}([x, t], r_e([x, t])) = |x|. \tag{7}$$

Proof. The deck action is by Euclidean isometries, so the infimum is independent of representatives and satisfies the triangle inequality. Only finitely many n can keep $|t - u - nL|$ bounded. Thus distance zero forces one deck translate to coincide with the other representative. Also the inverse image of each quotient ball is a union of Euclidean balls in the relative strip. Conversely a deck-saturated open set contains a ball about every representative, so its image is metric-open. This verifies the topology claim. The fundamental closed rectangle $[-A, A] \times [0, L]$ maps onto Q_e, proving compactness. With both transverse coordinates zero, (6) reduces to the usual circle distance. The formula for r_e respects the deck relation, and the vertical circle distance is bounded by each term in (6); hence the retraction is continuous. Setting $y = 0, u = t$ proves (7). □

Figure 1 illustrates the ambient identifications and retraction: the colored seam markers specify the gluing, and the squares are ambient points moved to the central circle, not zeta-zero data.

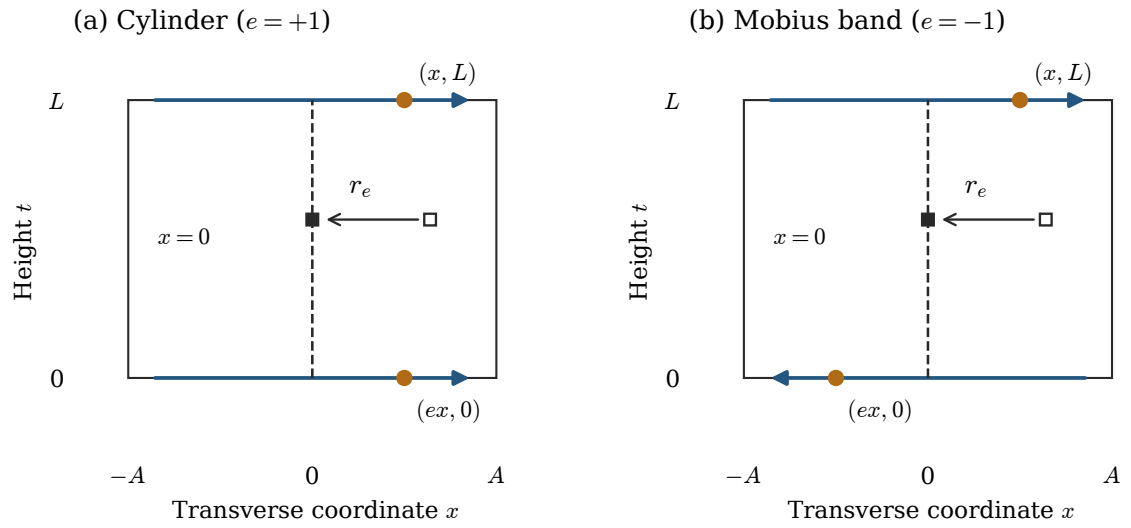


Figure 1: Ambient cylinder and Möbius fundamental domains. Colored circles represent the seam equivalence $(x, L) \sim (ex, 0)$; arrows encode equal or reversed transverse orientation. Squares illustrate the central retraction r_e. These are schematic ambient points, not zeta-zero measurements and not a claim that the zero divisor descends. The dashed segment closes into the central circle.

The map q_e pushes a strip point to its folded position. Applying it to every positive-height zero gives the raw pushforward, of infinite total mass. The next display instead takes zeros through height H and divides by $N(H)$. Thus $\nu_{e,H}$ and $\lambda_{L,H}$ describe a zero chosen with multiplicity from that finite sample. Define them, for $N(H) > 0$,

$$\nu_{e,H} = \frac{1}{N(H)} \sum_{0<\gamma\le H} m_\rho \, \delta_{q_e(x_\rho,\gamma)}, \qquad \lambda_{L,H} = \frac{1}{N(H)} \sum_{0<\gamma\le H} m_\rho \, \delta_{\{\gamma/L\}}. \tag{8}$$

Write $\mu_e = (j_e)_* m_L$, the uniform Haar probability on the central circle, viewed inside the quotient. On a compact metric space define

$$d_{\mathrm{Lip}}(\nu, \mu) = \sup_{\mathrm{Lip}(f)\le 1} \left| \int f \, \mathrm{d}\nu - \int f \, \mathrm{d}\mu \right|,$$

where tests are real-valued. Constants cancel because both masses are one. Here weak convergence means that the averages of every fixed continuous test function converge to its average under μ_e; it does not mean that any individual zero moves toward the circle.

Theorem 3.2. *For either deck sign, $\nu_{e,H}$ converges weakly to μ_e, unconditionally. More precisely,*

$$\begin{aligned} d_{\mathrm{Lip}}(\nu_{e,H}, \mu_e) &\le \frac{1}{N(H)} \sum_{0<\gamma\le H} m_\rho |x_\rho| + L D^*_{1/L}(H) \\ &= O_L\!\left(\frac{\log\log H}{\log H} \right). \end{aligned} \tag{9}$$

The weak-limit assertion only requires Hlawka equidistribution and (3); the displayed rate also uses (2).

Proof. Critical-line reflection preserves multiplicity at each positive height. Layer cake and (3), with its uniform constant C, give

$$\sum_{0<\gamma\le H} m_\rho |x_\rho| = \int_0^{1/2} \sum_{\substack{0<\gamma\le H \\ |x_\rho|>v}} m_\rho \, \mathrm{d}v$$
$$\le 2CH \log H \int_0^{1/2} e^{-v \log H/4} \, \mathrm{d}v \le 8CH.$$

The two-sided convention in the density input only enlarges this bound. By (7), replacing $f(q_e(x_\rho, \gamma))$ by $f(j_e([\gamma]_L))$ costs at most the first term of (9). The function $g(u) = f(j_e([Lu]))$ on $[0, 1]$ is periodic and L-Lipschitz. It is absolutely continuous. Stieltjes integration by parts, with $F_H(u) = \lambda_{L,H}([0, u))$, gives

$$\left| \int g \, \mathrm{d}(\lambda_{L,H} - \mathrm{d}u) \right| \le \sup_u |F_H(u) - u| \int_0^1 |g'(u)| \, \mathrm{d}u \le L D^*_{1/L}(H).$$

Periodic endpoints cancel; left and right CDF conventions differ only at atoms and have the same integral against g'. This proves the inequality. Now use (1) and (2) for the rate.

For completeness, weak convergence for every continuous complex-valued test does not require a transport duality theorem. Uniform continuity on Q_e makes its retraction error uniformly small on $|x| < \epsilon$. The exceptional proportion is $O(H^{-\epsilon/4})$ by (3) and (1). The restriction to the central circle is handled by Hlawka. Letting ϵ decrease after taking the limit proves the claim. □

Proposition 3.3. *The raw positive-height pushforward $\mathcal{D}_e = \sum_{\gamma>0} m_\rho \delta_{q_e(x_\rho,\gamma)}$ has infinite mass in every neighborhood of every central-circle point. It is not locally finite and is not a Radon measure.*

Proof. Such a neighborhood contains the image of a narrow transverse strip $|x| < \delta$ above a nonempty open height arc I. The metric formula justifies this statement even across the twisted seam. Through height H its mass is at least the count in (5) minus the count with $|x| \ge \delta$. The former is $(|I|/L + o(1))N(H)$; the latter is $o(N(H))$. Their difference diverges. A Radon measure is finite on compact sets, whereas Q_e is compact and already has infinite mass. □

No analogous conclusion is asserted at every off-center point: under RH, for example, an off-center neighborhood would contain no zeros. Theorem 3.2 says only that the weighted exceptional proportion vanishes; it does not put each zero on the central circle.

4 Critical-line residue complexity

The all-zero limit is informative, but it should not be quietly reassigned to a selected subsequence. We therefore isolate the weaker consequence available from the simple-zero proportion. Let Γ_+ be the distinct positive ordinates of all critical-line zeros, and let $R_L(H)$ count residues modulo L represented by *simple* critical-line zeros through height H.

Theorem 4.1. *For every fixed $L > 0$,*

$$R_L(H) \ge \frac{N_0^*(H)}{\lfloor H/L \rfloor + 1}, \qquad \liminf_{H\to\infty} \frac{R_L(H)}{\log H} \ge \frac{0.4L}{2\pi}. \tag{10}$$

In particular Γ_+ has no eventual support period.

Proof. Simple critical-line zeros have distinct ordinates. A residue class modulo L contains at most $\lfloor H/L \rfloor + 1$ positive heights through H, proving the first inequality. By (4), eventually $N_0^*(H) \geq 0.4N(H)$. Divide by $\log H$ and use $N(H)/(H \log H) \to 1/(2\pi)$ and $(\lfloor H/L \rfloor + 1)/H \to 1/L$. An eventual period would leave only finitely many residues, contradicting their unbounded number. □

Proposition 4.2. *Let K_L be the closure in $\mathbb{T}_L$ of the projected positive critical-line ordinates, or of any fixed tail of them. Then $m_L(K_L) \geq 0.4$.*

Proof. The all-zero height probability assigns K_L at least the proportion of simple critical-line zeros, apart from a finite prefix. Its limiting lower mass is at least 0.4 by (4). Since K_L is closed, weak convergence to Haar and the closed-set inequality imply that its Haar mass is at least that number, explicitly

$$0.4 \leq \liminf_{H\to\infty} \lambda_{L,H}(K_L) \leq \limsup_{H\to\infty} \lambda_{L,H}(K_L) \leq m_L(K_L),$$

identifying the unit circle with $\mathbb{T}_L$ by $u \mapsto [Lu]_L$. □

This positive-measure statement and (10) are deliberately weaker than equidistribution or density of the critical-line-only sequence. Selecting a positive-proportion subsequence does not supply either conclusion.

5 Stability of the observations

The next elementary estimate explains the sparse example. In a probability average, adding mass $M = o(N)$ changes a bounded test average by a vanishing amount. For a centered statistic divided by H, the corresponding condition is $M = o(H)$. Keeping these two scales separate explains which observations can overlook an added divisor.

Lemma 5.1. *Let D be a finite positive atomic measure of mass $N > 0$, and E another of mass $M \geq 0$. For every bounded measurable test h on their common target space,*

$$\left| \frac{\int h \,\mathrm{d}(D+E)}{N+M} - \frac{\int h \,\mathrm{d}D}{N} \right| \leq \frac{2\|h\|_\infty M}{N+M}. \tag{11}$$

For a reference probability m and $H > 0$,

$$\left| \frac{\int h \,\mathrm{d}E - M \int h \,\mathrm{d}m}{H} \right| \leq \frac{2\|h\|_\infty M}{H}. \tag{12}$$

Proof. Put $A = \int h \,\mathrm{d}D$, $B = \int h \,\mathrm{d}E$, and $I = \int h \,\mathrm{d}m$. The numerator in (11), after putting it over the denominator $N(N+M)$, is $NB - MA$. Use $|A| \leq \|h\|_\infty N$ and $|B| \leq \|h\|_\infty M$. The other inequality follows from $|B - MI| \leq |B| + M|I|$. □

For a zero divisor D define the smooth secondary functional on $\mathbb{T}_1$ by

$$\mathcal{S}^D_{\alpha,H}(h) = \frac{1}{H} \left(\sum_{0<\gamma\leq H} m_\rho h(\{\alpha\gamma\}) - N_D(H) \int_0^1 h(u) \,\mathrm{d}u \right). \tag{13}$$

Ford–Zaharescu [6, Theorem 1 and Corollary 2] prove for $h \in C^2(\mathbb{T}_1)$ that this converges to $\int h g_\alpha$ for zeta. For $\alpha > 0$ the density is zero unless $\alpha = r \log p/(2\pi q)$ for a prime p and coprime positive integers r, q. In the latter case

$$g_\alpha(u) = -\frac{\log p}{\pi} \operatorname{Re} \sum_{k\geq 1} p^{-rk/2} e^{-2\pi i q k u}. \tag{14}$$

The series is absolutely convergent. The theorem concerns each fixed C^2 test h: it is a smooth-test secondary limit, not weak convergence against every continuous test of the unbounded signed measures and not an established interval-indicator theorem. In the example below, the added mass is $O(\log H)$, so the stability bound at scale $1/H$ tends to zero. That bound also holds uniformly over test families with a common supremum-norm bound; the theorem from [6] supplies the base limit for each fixed smooth test.

6 An infinite off-critical perturbation

Fix $0 < \eta < 1/2$ and put $z_n = \eta + i2^n$ for $n \geq 1$. Define

$$P(w) = \prod_{n\geq 1} A_n(w), \qquad A_n(w) = (1 - w^2/z_n^2)(1 - w^2/\bar{z}_n^2). \tag{15}$$

This is a standard prescribed-divisor construction, used here as a worked example in the setting of [1]. The exponential heights make the added count $O(\log H)$; no arithmetic significance of the base two is claimed. The roots to be prescribed belong to P, not to the observed zeta divisor.

Theorem 6.1. *The product is an even real entire function with simple zeros exactly* $\{\pm\eta \pm i2^n : n \geq 1\}$. *It is positive on the real and imaginary axes. For* $|w| \leq R$,

$$\sum_{n>N} |A_n(w) - 1| \leq B_N(R) = \frac{2R^2}{3}4^{-N} + \frac{R^4}{15}16^{-N}. \tag{16}$$

If $B_N(R) < 1$, *its tail factor satisfies*

$$\left|\prod_{n>N} A_n(w) - 1\right| \leq \frac{B_N(R)}{1 - B_N(R)}. \tag{17}$$

Furthermore $\log \max_{|w|\leq R} |P(w)| = O((\log(R+2))^2)$.

Proof. Because $|z_n| \geq 2^n$, expansion of the paired factor gives

$$|A_n(w) - 1| \leq 2R^2 4^{-n} + R^4 16^{-n}.$$

Its compact-uniform summability proves normal convergence of the product, hence entire holomorphy. At a point outside the prescribed zeros, the tail has a nonzero limit by the summable-deviation infinite-product criterion; equivalently take a sufficiently small tail and use its convergent logarithm. Near a prescribed zero, all the other factors form a nonvanishing analytic function. The four roots of each factor are distinct and different factors have distinct heights. This proves the exact simple divisor.

Evenness and conjugation symmetry hold factorwise. For real u, v,

$$A_n(u) = |1 - u^2/z_n^2|^2, \qquad A_n(iv) = |1 + v^2/z_n^2|^2.$$

They are positive because z_n^2 is nonreal. Their infinite product cannot vanish there, by the criterion just proved, so the limit is positive.

Summing the two geometric tails gives (16). For a finite set of tail factors put $u_n = |A_n(w) - 1|$. Expanding the product and taking absolute values bounds its difference from one by the sum of its elementary symmetric polynomials in the u_n. The degree-k sum is at most $(\sum u_n)^k$, so the difference is at most $\sum_{k\geq 1} B_N^k = B_N/(1 - B_N)$. Pass to the product limit to obtain (17). This is a bound on the tail factor, even where the finite prefix is zero; no division by that prefix is required.

Finally $|A_n(w)| \le (1 + R^2 4^{-n})^2$. Split the sum of its logarithms at $J = \lceil \log_2 \max(2, R) \rceil$. The first J terms are individually $O(\log(R+2))$, and the remaining logarithms have bounded sum using $\log(1+v) \le v$. This proves the maximum-modulus bound. In particular P has entire-function order zero. □

Figure 2 is a reading aid: its left panel shows the positive-height pairs of prescribed roots; their conjugates complete the fourfold set at negative heights. Its right panel shows why their positive-height count is only logarithmic in H.

Theorem 6.2. *Let $X(s) = \xi(s)P(s - 1/2)$. It has infinitely many prescribed off-critical zeros in the open strip, satisfies $X(1-s) = X(s)$ and the real conjugation symmetry, and retains every critical-line zero with its original order. Its real critical-line restriction has the same sign away from zeros. For $H \ge 2$,*

$$N_X(H) = N(H) + M(H), \qquad M(H) = 2\lfloor \log_2 H \rfloor. \tag{18}$$

It has the same cylinder and Möbius probability limits as zeta, and

$$\mathcal{S}^X_{\alpha,H}(h) - \mathcal{S}^{\xi}_{\alpha,H}(h) \to 0 \tag{19}$$

for every fixed α and bounded measurable test h. Consequently all the proved smooth limits (14) persist.

Proof. Divisors of nonzero holomorphic products add, including at accidental coincidences with xi zeros. Theorem 6.1 inserts exactly two positive-height roots for each $2^n \le H$, proving (18) with the closed endpoint convention. The symmetry and sign assertions follow from the even real structure and the positive imaginary-axis values of P. Exact function values on the critical line are not preserved. Since $M(H) = O(\log H) = o(N(H))$, (11) transfers the quotient probability limits. Inequality (12) bounds the difference in (19) by $2\|h\|_\infty M(H)/H = O(\|h\|_\infty \log H/H)$. This transfers a limit whenever it exists; it does not establish an unproved interval limit for either function. □

The construction now marks the distinction between an added divisor and the actual zeta divisor. It has extra off-critical zeros, even though the stated normalized and fixed-smooth-test observations agree asymptotically. The Riemann–von Mangoldt main terms and its $O(\log H)$ remainder class are unchanged. The density bound (3) is unchanged as a big-O class too: the additional two-sided count is $O(\log H)$, absorbed uniformly since $1 - (\sigma - 1/2)/4 \ge 7/8$ on the stated range. The quotient rate is likewise unchanged in order by (11).

Figure 3 displays two explicit bounds used in the construction.

Proposition 6.3. *For every fixed $x > 1$, the leading term and error class in Landau's formula*

$$\sum_{0<\gamma\le H} m_\rho x^\rho = -\frac{H}{2\pi}\Lambda(x) + O_x(\log H)$$

are preserved by X. Here $\Lambda(x)$ is zero unless x is an integral prime power. The error class of the uniform version in [6, Lemma 1] also absorbs the perturbation for $x > 1, H \ge 2$.

Proof. Every added zero has real part between zero and one, so its contribution has modulus at most x. The total difference is at most $xM(H) = O(x \log H)$. For fixed x this gives the first claim. For varying x, the cited uniform error is $O(x \log^2(2xH) + \log(2H)/\log x)$. Its first component absorbs this bound. The nearest-prime-power main term is not changed. □

This assertion concerns asymptotic classes with unspecified constants. It does not preserve an exact Weil explicit formula or an Euler product. An exact added-divisor term remains present and can be detected by suitable finite-height tests.

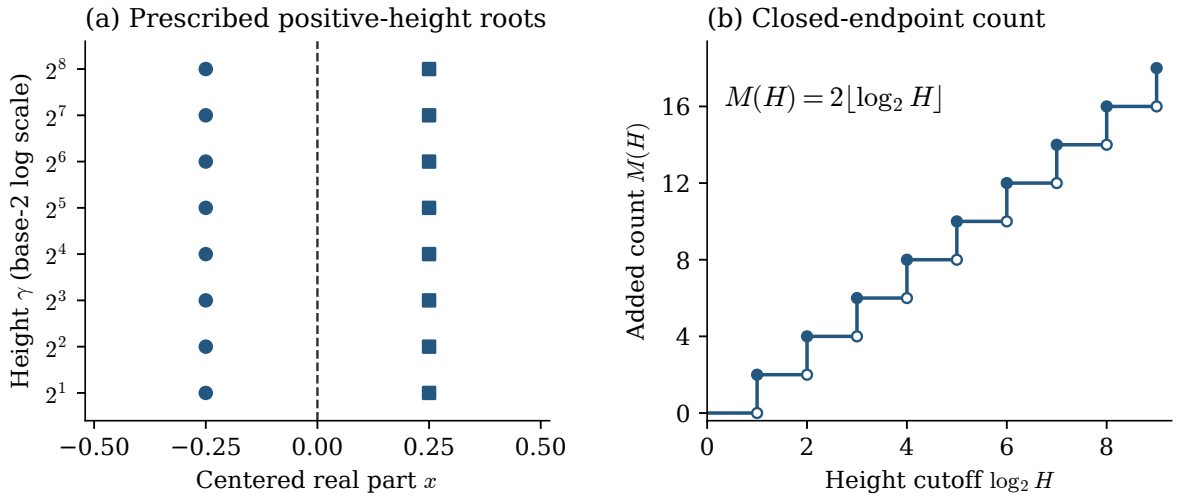


Figure 2: The prescribed perturbation for $\eta = 1/4$, not observed zeta zeros. Left: positive-height roots of P, for $1 \le n \le 8$, with the height axis on a base-two logarithmic scale. Conjugate roots at negative height are omitted. Right: the exact added count $M(H)$ through $H = 2^9$; filled dots include the dyadic endpoint and open dots show left limits. A coincidence with a xi zero adds multiplicity rather than deleting a root.

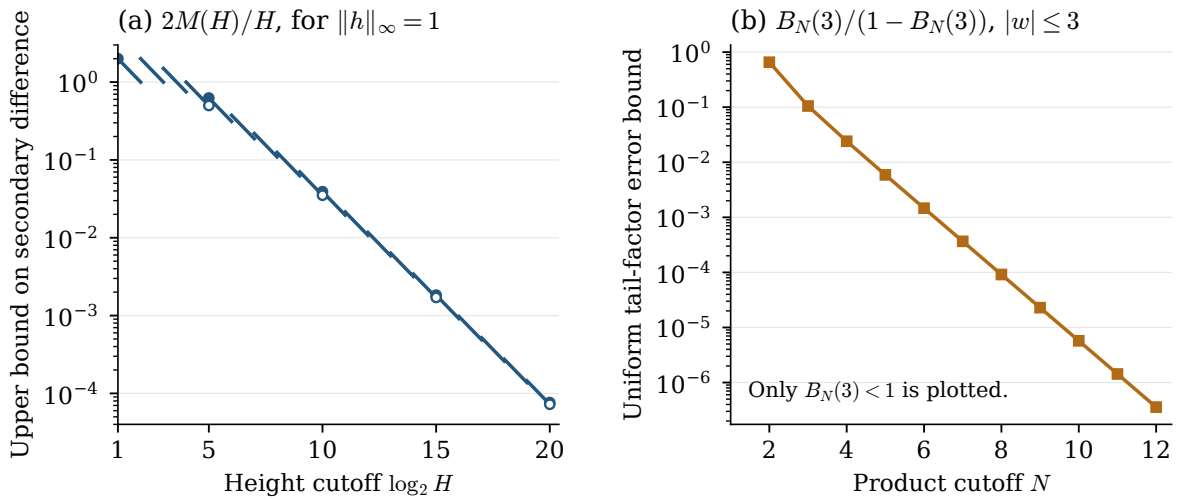


Figure 3: Formula-based upper bounds, not measured errors. Left: the secondary-statistic difference bound $2M(H)/H$ for $\|h\|_\infty = 1$, $2 \le H \le 2^{20}$; open and filled dots show selected left limits and included endpoints. Right: the compact tail-factor bound $B_N(3)/(1 - B_N(3))$ on $|w| \le 3$, plotted only when $B_N(3) < 1$. The right-hand vertical axis is logarithmic, as is the left-hand vertical axis. Exact rational source rows and replay code accompany the figures; the universal bounds are proved in the text.

7 The arithmetic boundary: an ordinary Dirichlet series

The preceding construction can also match the pole normalization, so the geometric and averaged tests alone have not identified zeta. We now ask for an ordinary Dirichlet-series representation $\sum_{n\geq 1} a_n n^{-s}$ that converges absolutely in some right half-plane. The next proposition gives a direct arithmetic condition distinguishing the displayed construction from zeta: its normalized multiplier prevents such a representation. Put

$$\widetilde{\zeta}(s) = \zeta(s)\frac{P(s-1/2)}{P(1/2)}. \tag{20}$$

The denominator is a strictly positive constant.

Proposition 7.1. *The meromorphic function $\widetilde{\zeta}$ has the same gamma-factor functional equation as zeta, the same pole at one with residue one, and the same trivial-zero multiplicities. Nevertheless it is not representable by any ordinary Dirichlet series absolutely convergent on a right half-plane.*

Proof. The multiplier is entire, nonzero on the real axis, and invariant under $s \mapsto 1-s$. Multiplying zeta's meromorphic functional equation therefore gives the same equation, with its usual gamma factor. Its value at $s = 1$ is one, proving the residue statement. It is nonzero at each negative even integer, so it retains every trivial-zero multiplicity.

For real u, direct expansion gives

$$A_n(u) = 1 + \frac{2(4^n - \eta^2)}{(\eta^2 + 4^n)^2}u^2 + \frac{u^4}{(\eta^2 + 4^n)^2} \geq 1.$$

Thus $P(u) \geq A_1(u) \to \infty$ as $u \to \infty$. Since $\zeta(\sigma) \to 1$ on the positive real axis, (20) tends to infinity there. In contrast, if $F(s) = \sum_{n\geq 1} a_n n^{-s}$ converges absolutely at a real σ_0, dominated convergence gives $F(\sigma) \to a_1$ as $\sigma \to \infty$: for $\sigma \geq \sigma_0$, the summands are dominated by the summable $|a_n|n^{-\sigma_0}$. This contradiction proves the claim. □

The role of an ordinary Dirichlet series is consonant with Hamburger's converse theorem, whose precise classical formulation is recalled in [8, Introduction]: if two ordinary Dirichlet series f, g converge absolutely for $\operatorname{Re} s > 1$, if $(s-1)f(s)$ and $(s-1)g(s)$ are entire of finite order, and if

$$\pi^{-s/2}\Gamma(s/2)f(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)g(1-s),$$

then $f = g = c\zeta$. We do not invoke that theorem to prove the proposition; the positive-real-axis obstruction suffices. Extensions of converse rigidity are treated by Burnol [3]; related explicit non-Dirichlet functions occur in [9]. The zero-insertion literature [2, 1] is a closer predecessor to our construction. This proposition is an elementary check for the displayed multiplier, not a new general nonuniqueness theorem.

8 Verification, interpretation, and open problems

The exact verifiers supplied with this draft check the independent geometric tails in (16), finite complex paired-factor and tail deviations, prescribed roots, axis-positivity samples, dyadic count endpoints, and the fail-closed assertion guards. They are finite arithmetic checks; holomorphy of the infinite product, its zero divisor, and the zeta inputs remain matters of the written arguments. The real-test algebra in Lemma 5.1 is also kernel-checked in the companion `EmpiricalStability.lean` against Lean 4.33.1 and a pinned Mathlib checkout. Its reported logical axioms are `propext`, `Classical.choice`, and `Quot.sound`. This is limited

coverage, not a Lean proof of the analytic input theorems or of the whole manuscript. Earlier coordinate and zeta-support modules retain their separate coverage. The figure companion provides exact rational source rows, vector/raster exports, independent formula checks, and regressions detecting a wrong normalization or dyadic endpoint; it uses no empirical zeta-zero dataset.

The narrative has three steps. An ambient height quotient always exists, but descent of the original locally finite divisor would require deck compatibility, which the support theorem excludes. Normalized finite-height probabilities nevertheless converge to the central-circle Haar probability. Finally, an infinite sparse off-critical divisor can retain that limit and the stated fixed-smooth secondary signatures. The half-twist changes the ambient topology without recovering the information lost by these observations; the ordinary Dirichlet-series condition is one concrete arithmetic discriminator for the displayed construction.

The two bounds in Lemma 5.1 offer a concrete starting point for further study: what change of scale or growth of test norms would let an observation detect the added sparse mass? The aim is to retain a useful limiting theory while making that information visible. Related questions concern stronger information about the projected critical-line-only subsequence and which exact arithmetic identities distinguish zeta from sparse perturbations. The interval secondary asymptotics conjectured in [6, 7] are not settled here.

References

[1] P. M. Gauthier and X. Xarles, *Perturbations of L-functions with or without non-trivial zeros off the critical line*, arXiv:0911.5135v1 (2009), especially Proposition 1 and Theorem 2.

[2] P. M. Gauthier and E. S. Zeron, *Small Perturbations of the Riemann Zeta Function and their Zeros*, Comput. Methods Funct. Theory 4 (2004), 143–150; doi:10.1007/BF03321061.

[3] J.-F. Burnol, *Deux extensions de Théorèmes de Hamburger*, arXiv:1106.4749 (2011).

[4] NIST Digital Library of Mathematical Functions, Section 25.4, equations 25.4.3–25.4.4.

[5] H. M. Bui, J. B. Conrey, and M. P. Young, *More than 41% of the zeros of the zeta function are on the critical line*, Acta Arith. 150 (2011), 35–64; arXiv:1002.4127v2.

[6] K. Ford and A. Zaharescu, *On the distribution of imaginary parts of zeros of the Riemann zeta function*, arXiv:math/0405459v2 (2004), especially the Introduction, Theorem 1, Corollary 2, and Lemmas 1–2.

[7] K. Ford, K. Soundararajan, and A. Zaharescu, *On the distribution of imaginary parts of zeros of the Riemann zeta function, II*, arXiv:0805.2745v3 (2009).

[8] J. Kaczorowski, G. Molteni, and A. Perelli, *A converse theorem for Dirichlet L-functions*, Comment. Math. Helv. 85 (2010), 463–483; author-hosted text.

[9] T. Nakamura, *L-functions with the Lindelöf and Riemann hypotheses and trivial zeros at even negative integers*, Ann. Mat. Pura Appl. (2026), doi:10.1007/s10231-026-01726-9.